ADCS Propagator

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1. Introduction

This report presents the implementation of an attitude propagator for satellites developed in Python. The model excludes the six degrees of freedom associated with position and velocity, focusing solely on the six degrees of freedom that describe the satellite's attitude.

The implementation follows a modular approach, allowing for easy reuse of functions and packages in future projects.

This document does not go into excessive theoretical detail regarding the ADCS subsystem, as it is intended for readers already familiar with the topic.

2. Initial Considerations

Some preliminary considerations related to the implementation of the attitude propagator are presented below.

- The chosen state vector is defined as:

$$\boldsymbol{y} = \begin{bmatrix} \boldsymbol{q} \\ \boldsymbol{\omega} \end{bmatrix}, \tag{1}$$

  where $\boldsymbol{q}$ represents the quaternion and $\boldsymbol{\omega}$ the angular velocity. The quaternion is defined such that its scalar part is the last component, q_4. The scalar part is denoted as η, and the vector part as $\boldsymbol{\epsilon}$. Quaternions are chosen instead of Euler angles for convenience and to avoid the singularities inherent to the latter, which could lead to numerical issues during propagation.

- The notation used for angular velocities is $\boldsymbol{\omega}^{\gamma}_{\alpha\beta}$, which is read as "angular velocity of frame α with respect to frame β, expressed in frame γ."

- A Direction Cosine Matrix (DCM) is denoted as $\boldsymbol{C}_{\beta\alpha}$, which represents the transformation of a vector expressed in frame α into its components in frame β. This same approach is taken for a quaternion $\boldsymbol{q}_{\beta\alpha}$.

- The derivative of the state vector is modeled according to:

$$\dot{\boldsymbol{\epsilon}} = \frac{1}{2}(\eta \boldsymbol{I} + \boldsymbol{\epsilon}^{\times})\boldsymbol{\omega}_{21}, \tag{2}$$

$$\dot{\eta} = -\frac{1}{2}\boldsymbol{\epsilon}^{T}\boldsymbol{\omega}_{21}, \tag{3}$$

  while the angular velocity is modeled according to Euler's equation for a rigid body, written in the body fixed frame, b, with respect to an inertial frame, G, so that the inertia tensor remains constant:

$$\dot{\boldsymbol{\omega}}^{b}_{bG} = \boldsymbol{I}_b^{-1}\left(\boldsymbol{T} - \left(\boldsymbol{\omega}^{b}_{bG}\right)^{\times}\boldsymbol{I}_b\boldsymbol{\omega}^{b}_{bG}\right), \tag{4}$$

  where $\boldsymbol{T}$ represents the external torques applied to the satellite, and $\boldsymbol{I}$ the inertia tensor of the satellite, both expressed in the body fixed frame.

- All rotations are considered positive following the right-hand rule convention.

- Although quaternions are used as the main attitude representation in the state vector, additional functions are implemented to obtain the Euler angles, which provide a more intuitive visualization. The chosen rotation sequence is 3(z)-2(y)-1(x), where rotation around the z-axis corresponds to yaw (ψ), around the y-axis to pitch (θ), and around the x-axis to roll (ϕ), as shown in Figure 1.

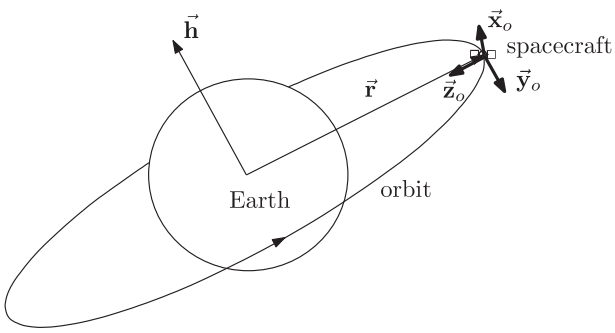

Figure 1: Orbiting reference frame definition.[1]

- The numerical scheme used for propagation is the fourth-order Runge–Kutta

(RK4) method. During my Master's Thesis [2], the results obtained using RK4 were compared with GMAT orbital propagation results, showing differences on the order of millimeters when compared to the RK89 integrator, as shown in Figure 3.

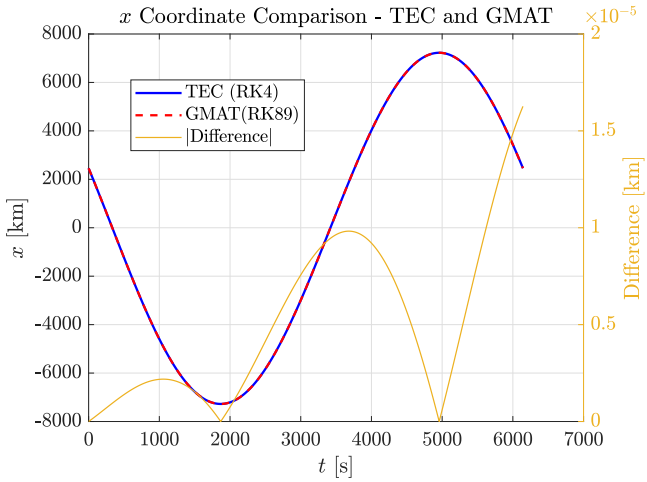

Figure 2: Comparison between RK4 from orbital propagator (TEC) and GMAT RK89 for same time step and initial conditions.[2]

3. Results

3.1. Sensor

The sensor establishes a relationship between its own reference frame, denoted as s, and the inertial frame, denoted as i, through the quaternion:

$$\boldsymbol{q}_{si} = \begin{bmatrix} 0.936680 \\ 0.216195 \\ 0.124597 \\ 0.245695 \end{bmatrix} \tag{5}$$

Hence, the transformation between the sensor frame s and the inertial frame i is known. To obtain the relationship between the body frame b and the inertial frame i, it is sufficient to relate the body frame to the sensor frame. In terms of DCMs, this relationship is expressed as:

$$\boldsymbol{C}_{bi} = \boldsymbol{C}_{bs}\,\boldsymbol{C}_{si}. \tag{6}$$

Given the rotation matrices $\boldsymbol{C}_{ts} = \boldsymbol{C}(\boldsymbol{\epsilon}_1, \eta_1)$, $\boldsymbol{C}_{bt} = \boldsymbol{C}(\boldsymbol{\epsilon}_2, \eta_2)$, and $\boldsymbol{C}_{bs} = \boldsymbol{C}(\boldsymbol{\epsilon}_3, \eta_3)$, and knowing that:

$$\boldsymbol{C}_{bs} = \boldsymbol{C}_{bt}\,\boldsymbol{C}_{ts}, \tag{7}$$

the associated quaternions are related by:

$$\boldsymbol{\epsilon}_3 = \eta_2\boldsymbol{\epsilon}_1 + \eta_1\boldsymbol{\epsilon}_2 + \boldsymbol{\epsilon}_1^{\times}\boldsymbol{\epsilon}_2, \tag{8}$$

$$\eta_3 = \eta_1\eta_2 - \boldsymbol{\epsilon}_2^T\boldsymbol{\epsilon}_1. \tag{9}$$

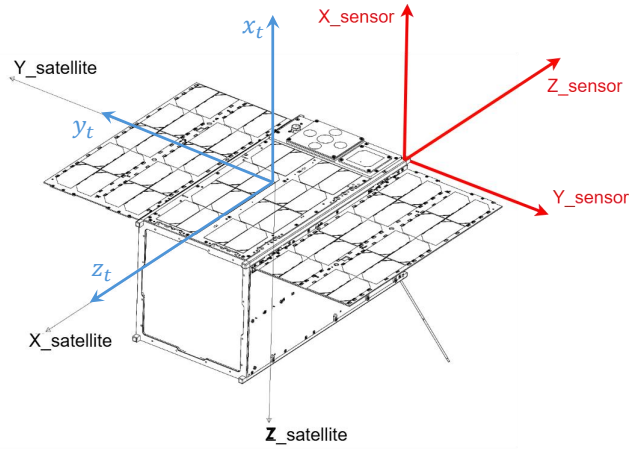

Figure 3: Reference frames involved.

Table 1: Rotations used to obtain the quaternion relating the sensor frame s to the body frame b, $\boldsymbol{q}_{bs}$.

Rotation order	Angle [deg]	Axis
1	180	$\boldsymbol{x}_{\text{sensor}}$
2	-90	$\boldsymbol{y_t}$

In this way, the resulting transformation is:

$$\boldsymbol{C}_{bs} = \begin{bmatrix} 0 & 0 & -1 \\ 0 & -1 & 0 \\ -1 & 0 & 0 \end{bmatrix}, \tag{10}$$

which is equivalent to the quaternion:

$$\boldsymbol{q}_{bs} = \begin{bmatrix} 0.70710 \\ 0 \\ -0.70710 \\ 0 \end{bmatrix}. \tag{11}$$

Taking into account Equation 6, the orientation of the body frame b with respect to the inertial frame i is given by:

$$\boldsymbol{C}_{bi} = \begin{bmatrix} -0.3397 & 0.4064 & 0.8482 \\ -0.3438 & 0.7858 & -0.5141 \\ -0.8755 & -0.4662 & -0.1272 \end{bmatrix}, \quad (12)$$

which corresponds to the equivalent quaternion:

$$\boldsymbol{q}_{bi} = \begin{bmatrix} 0.0209 \\ 0.7504 \\ -0.3266 \\ -0.5742 \end{bmatrix}. \quad (13)$$

The associated Euler angles are:

$$\psi = 129.89°, \quad (14)$$

$$\theta = -58.02°, \quad (15)$$

$$\phi = -103.89°. \quad (16)$$

3.2. Attitude Propagation

The initial conditions for the propagation are described below.

- The initial quaternion is:

$$\boldsymbol{q}_0 = \begin{bmatrix} 0.0209 \\ 0.7504 \\ -0.3266 \\ -0.5742 \end{bmatrix}. \quad (17)$$

- The initial angular velocity is:

$$\boldsymbol{\omega}^b_{bG0} = \begin{bmatrix} 1 \\ 2 \\ 0 \end{bmatrix} \text{deg/s}. \quad (18)$$

- The inertia tensor is:

$$\boldsymbol{I} = \begin{bmatrix} 1.0 & 0.1 & 0.1 \\ 0.1 & 2.0 & 0.1 \\ 0.1 & 0.1 & 0.3 \end{bmatrix} \text{kg·m}^2 \quad (19)$$

- The propagation time is $t = 1000$ s.
- The time step chosen is dt= 1 s.
- No external torques are included: $\boldsymbol{T} = \boldsymbol{0}$.

The results are shown in Figures 4, 5 and 6.

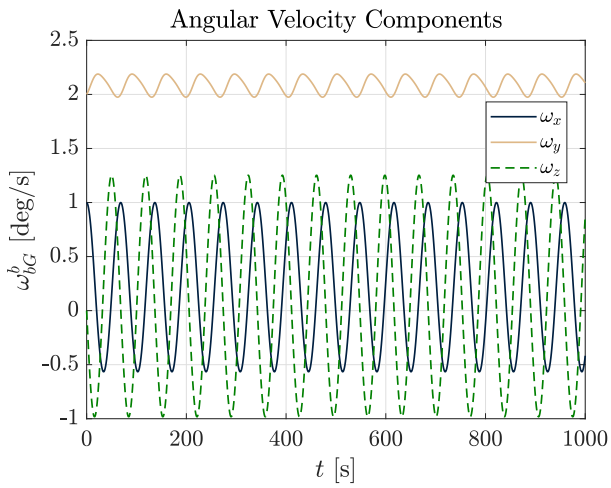

Figure 4: Angular velocity as a function of time for the non-perturbed case.

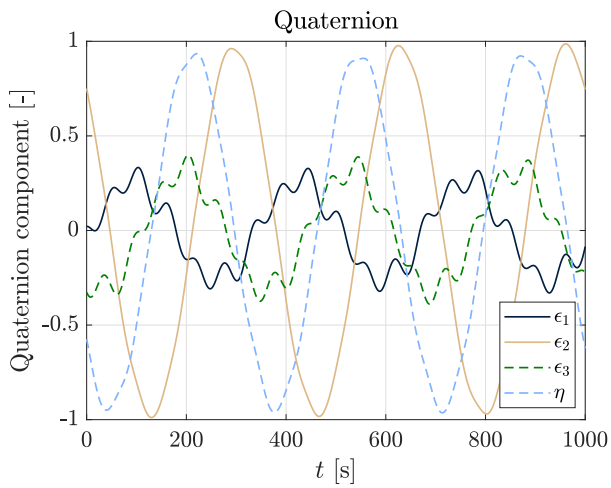

Figure 5: Quaternion components as a function of time for the non-perturbed case.

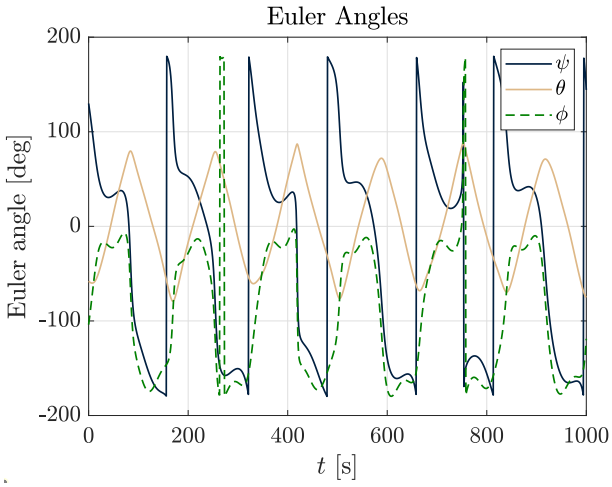

Figure 6: Euler angles as a function of time for the non-perturbed case.

Table 2: Components of the state vector for attitude at $t = 1000$ s.

State Component	Value
ϵ_1	-0.0851
ϵ_2	0.7449
ϵ_3	-0.2223
η	-0.6232
ω_x [deg/s]	-0.4178
ω_y [deg/s]	2.1067
ω_z [deg/s]	0.8753

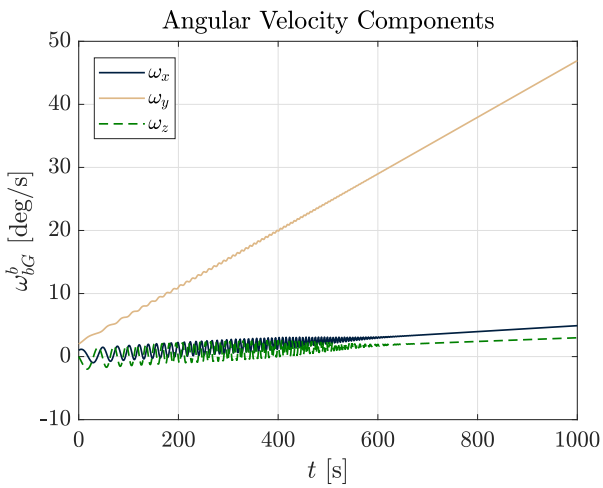

Figure 7: Angular velocity as a function of time for the perturbed case.

In the absence of external perturbations, the system remains stable over time, reaching a steady-state that is actually achieved from the very beginning of the simulation.

An external torque of $\boldsymbol{T} = [0.001, 0.0015, 0.0]^T$ N·m is now applied. Figures 7, 8, and 9 show the resulting system response under this perturbation from the same initial conditions.

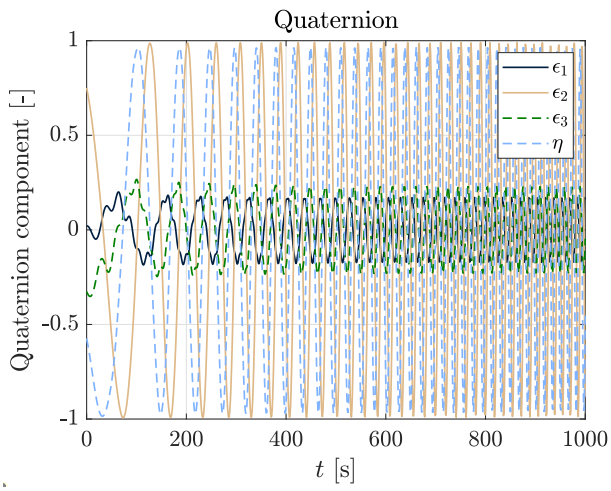

Figure 8: Quaternion components as a function of time for the perturbed case.

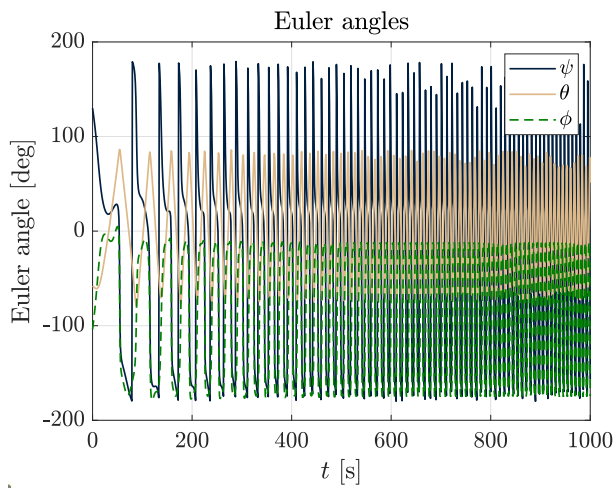

Figure 9: Euler angles as a function of time for the perturbed case.

The results show a continuous increase in all angular velocity components over time. This behavior is expected, as the presence of external torques continuously contributes to changes in

angular velocity in accordance with Equation 4, and no actuators are being modeled. Notably, the angular velocity about the body z-axis also increases over time, despite no external torque being applied along this axis. This occurs due to the coupling introduced by the satellite's inertia matrix. As a direct consequence, the growth in angular velocity leads to a larger variation in the Euler angles over time, as can be observed in Figure 9.

References

[1] Anton H.J De Ruiter, Christopher J. Damaren, and James R. Forbes. *Spacecraft Dynamics and Control.*

[2] Alejandro Fernández Ruiz. *Optimization of Deorbit Maneuvers and Reentry Conditions for LEO Capsules.*

[3] David A. Vallado. *Fundamental of Astrodynamics and Applications.*